

Chapter 3

Descriptive Statistics: Numerical Methods

- In this chapter we study numerical methods for describing the important aspects of a set of measurements.
- If the measurements are values of a **quantitative** variable, we often describe
 - (1) what a typical measurement might be, and
 - (2) how the measurements vary, or differ, from each other.
- The graphical displays of Chapter 2 and the numerical methods of this chapter give us a basic understanding of the important aspects of a set of measurements. We will illustrate this by continuing to analyse the car mileages, payment times, bottle design ratings, and cell phone usages (**if time permits**) introduced in Chapters 1 and 2.

Chapter Outline

3.1 Describing Central Tendency

3.2 Measures of Variation

3.3 Percentiles, Quartiles and Box-and-Whiskers Displays (学生自学)

3.1 Describing Central Tendency

- In addition to describing the shape of a distribution, want to describe the data set's central tendency
- A measure of central tendency represents the **center** or **middle** of the data
- May or may not be a typical value.

Parameters and Statistics

- Population mean is a **population parameter**.

A **population parameter** is a number calculated using the **population** measurements that describes some aspect of the population.

- Sample mean is a **Sample statistic**.

A **sample statistic** is a number calculated using the **sample** measurements that describes some aspect of the sample.

Point Estimates and Sample Statistics

A **point estimate** is a one-number estimate of the value of a **population** parameter.

A **sample statistic** is a number calculated using **sample** measurements that describes some aspect of the sample.

- The sample mean is a point estimate of the population mean.

- Use sample statistics as point estimates of the population parameters.

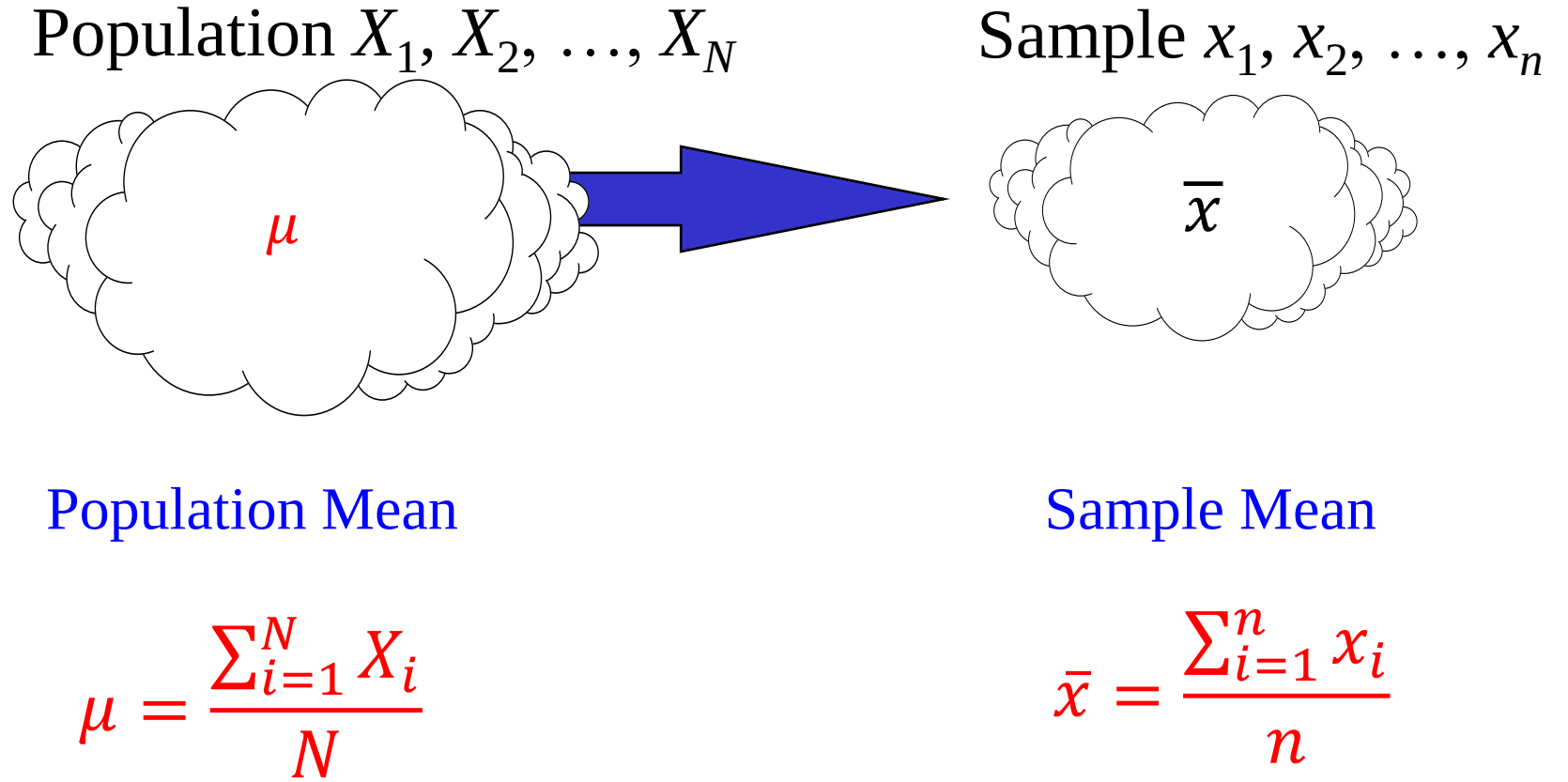
The **sample mean**, denoted $\bar{x}$, is a sample statistic and is the average of the sample measurements.

3.1.2 Measures of Central Tendency

- **Mean, μ** (pronounced mew): The average or expected value
- **Median, M_d** : The value of the middle point of the ordered measurements
- **Mode, M_o** : The most frequent value

The Mean

- Adding all the population measurements,
- and then dividing the resulting sum by the number of population measurements.



Example 3.1: Car Mileage Case

TABLE 3.1 A Sample of 50 Mileages  GasMiles

30.8	30.8	32.1	32.3	32.7
31.7	30.4	31.4	32.7	31.4
30.1	32.5	30.8	31.2	31.8
31.6	30.3	32.8	30.7	31.9
32.1	31.3	31.9	31.7	33.0
33.3	32.1	31.4	31.4	31.5
31.3	32.5	32.4	32.2	31.6
31.0	31.8	31.0	31.5	30.6
32.0	30.5	29.8	31.7	32.3
32.4	30.5	31.1	30.7	31.4

Example 3.1: Sample mean for first 5 car mileages from Table 3.1:

30.8, 31.7, 30.1, 31.6, 32.1

$$\bar{x} = \frac{\sum_{i=1}^5 x_i}{5} = \frac{x_1 + x_2 + x_3 + x_4 + x_5}{5}$$

$$\bar{x} = \frac{30.8 + 31.7 + 30.1 + 31.6 + 32.1}{5} = \frac{156.3}{5} = 31.26$$

Of course, intuitively, we are likely to obtain a more accurate point estimate of the population mean by using all of the available sample information. The sum of all 50 mileages can be verified to be

$$\sum_{i=1}^{50} x_i = x_1 + x_2 + \cdots + x_{50} = 30.8 + 31.7 + \cdots + 31.4 = 1578$$

Therefore, the mean of the sample of 50 mileages is

$$\bar{x} = \frac{\sum_{i=1}^{50} x_i}{50} = \frac{1578}{50} = 31.56$$

This point estimate says we estimate that the mean mileage that would be obtained by all of the new midsize cars that will or could potentially be produced this year is 31.56 mpg. Unless we are extremely lucky, however, there will be sampling error. That is, the point estimate $\bar{x} = 31.56$ mpg, which is the average of the sample of 50 randomly selected mileages, will probably not exactly equal the population mean μ , which is the average mileage that would be obtained by all cars. Therefore, although $\bar{x} = 31.56$ provides some evidence that μ is at least 31 and thus that the automaker should get the tax credit, it does not provide definitive evidence.

The Median

- Median is more meaningful than mean ?
- The median M_d is a value such that 50% of all measurements, after having been arranged in numerical order, lie above (or below) it.
 1. If the number of measurements is odd, the median is the middlemost measurement in the ordering.
 2. If the number of measurements is even, the median is the average of the two middlemost measurements in the ordering.

Example: Median

- Car Mileage Case: First five observations from Table 3.1:

30.8, 31.7, 30.1, 31.6, 32.1

In order: 30.1, 30.8, 31.6, 31.7, 32.1

There is an odd so median is one in middle, or 31.6

- Six exercise classes example

15, 30, 30, 34, 41, 60

Median is the average of the two in the middle or $(30 + 34) / 2 = 32$

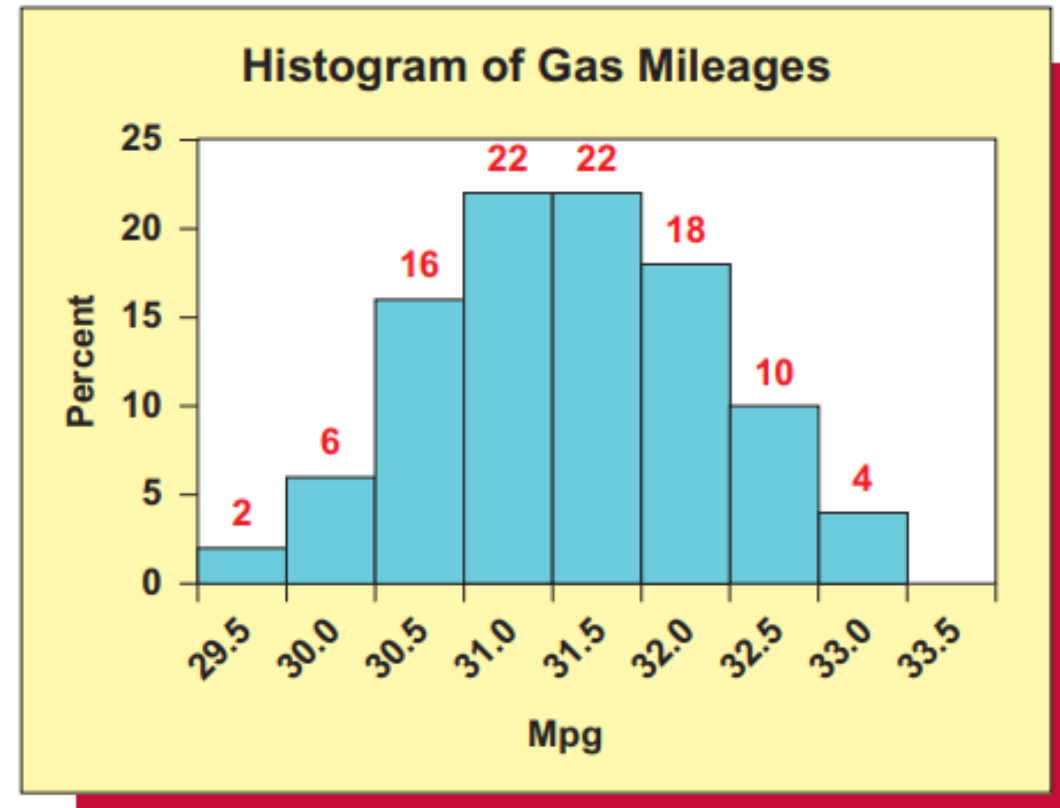
The Mode

- The mode M_0 of a population or sample of measurements is the measurement that occurs most frequently.
- Modes are the values that are observed “most typically”.
- Sometimes higher frequencies at two or more values
 - If there are two modes, the data is *bimodal*
 - If more than two modes, the data is *multimodal*
- When data are in classes, the class with the highest frequency is the modal class.

Example: Mode

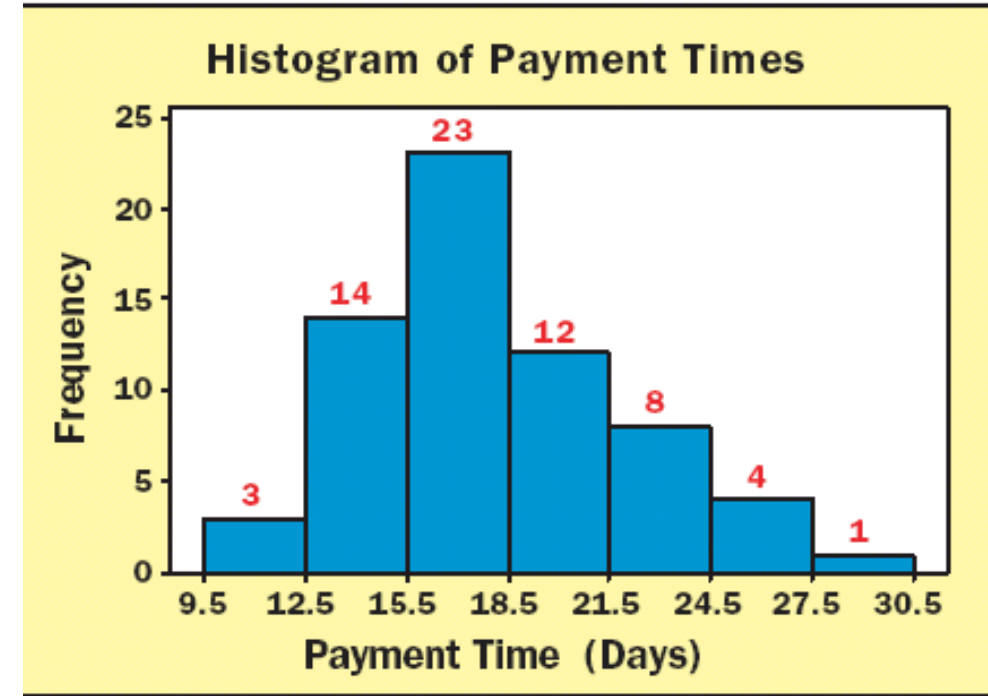
Figure 3.2 shows a histogram of the car mileages that has two car **modal classes** — the class from 31.0 mpg to 31.5 mpg and the class from 31.5 mpg to 32.0 mpg. Because the mileage 31.5 is in the middle of the modal classes, we might estimate that the population mode for the new car is 31.5 mpg.

FIGURE 3.2 A Percent Histogram Describing the 50 Mileages



The right figure is a histogram of **the e-Billing Case**.

- the class from 16 days to 19 days has highest frequency and therefore is the **modal class** of the e-Billing.



Relationships among Mean, Median and Mode

FIGURE 3.3 Typical Relationships among the Mean μ , the Median M_d , and the Mode M_o

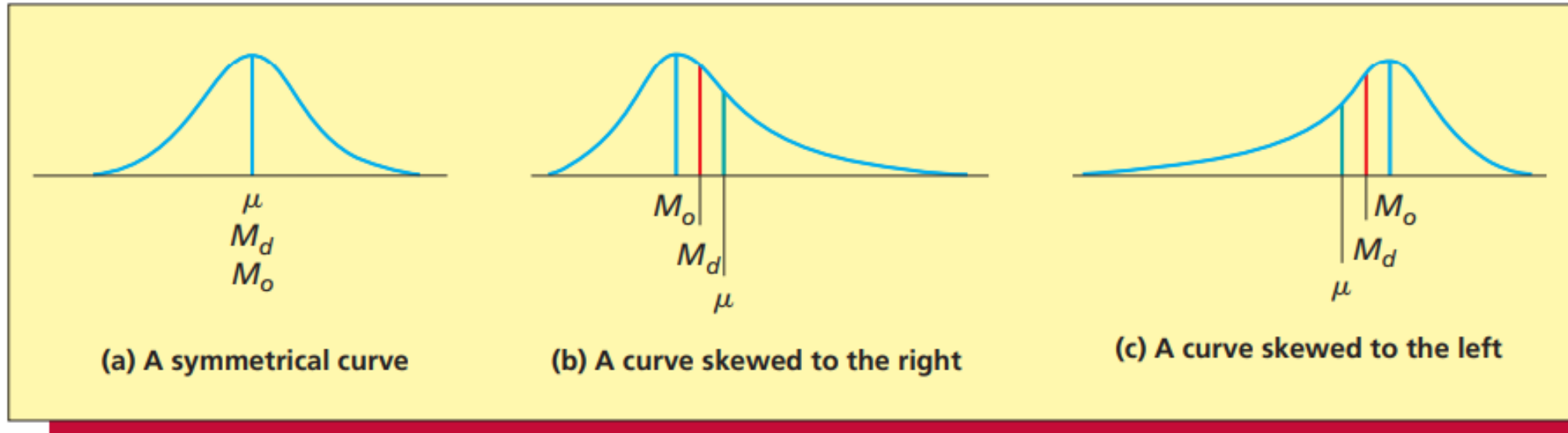


Figure 3.3(a) depicts a population described by a symmetrical relative frequency curve. When the frequency distribution of a sample of measurements is approximately symmetrical, then the sample mean, median, and mode will be nearly the same.

Figure 3.3(b) depicts a population that is skewed to the right. Here the population mean is larger than the population median, and the population median is larger than the population mode (the mode is located under the highest point of the relative frequency curve). To understand, we study the following examples.

Example 3.2: The distribution of 12 household incomes in a city is shown in dollars:

7,524	11,070	18,211	26,817	36,551	41,286
49,312	57,283	72,814	90,416	135,540	190,250

- The median is $(41,286 + 49,312) / 2 = 45,299$.
- The mean of the incomes is the sum of the incomes, 737,074, divided by 12, or \$61,423 (rounded to the nearest dollar). Here, **the mean has been affected by averaging in the large incomes** \$135,540 and \$190,250 and thus is larger than the median.
- **The median is resistant to large / small values** but **the mean is not**. For example, if the largest income were smaller—say \$150,000—the median would remain the same but the mean would decrease. If the largest income were larger—say \$300,000—the median would also remain the same but the mean would increase.
- Similarly, the median is resistant to values that are much smaller than most of the measurements. In general, we say that the median is resistant to **extreme values** (highest/smallest measurements).
- In general, **the median income is more representative of a typical income**.

Example 3.3: The Marketing Research Case: Rating A Bottle Design

60 bottle design ratings – original data is in Chapter 1 and repeated in next page.

Mean = 30.35. Median = 31.

Because the histogram of the bottle design ratings in Figure 3.5 is not highly skewed to the left, the sample mean is not much less than the sample median.

Therefore, using the mean as our measure of central tendency, we estimate that the mean rating of the new bottle design that would be given by all consumers is 30.35.

FIGURE 3.5 Excel Frequency Histogram of the 60 Bottle Design Ratings

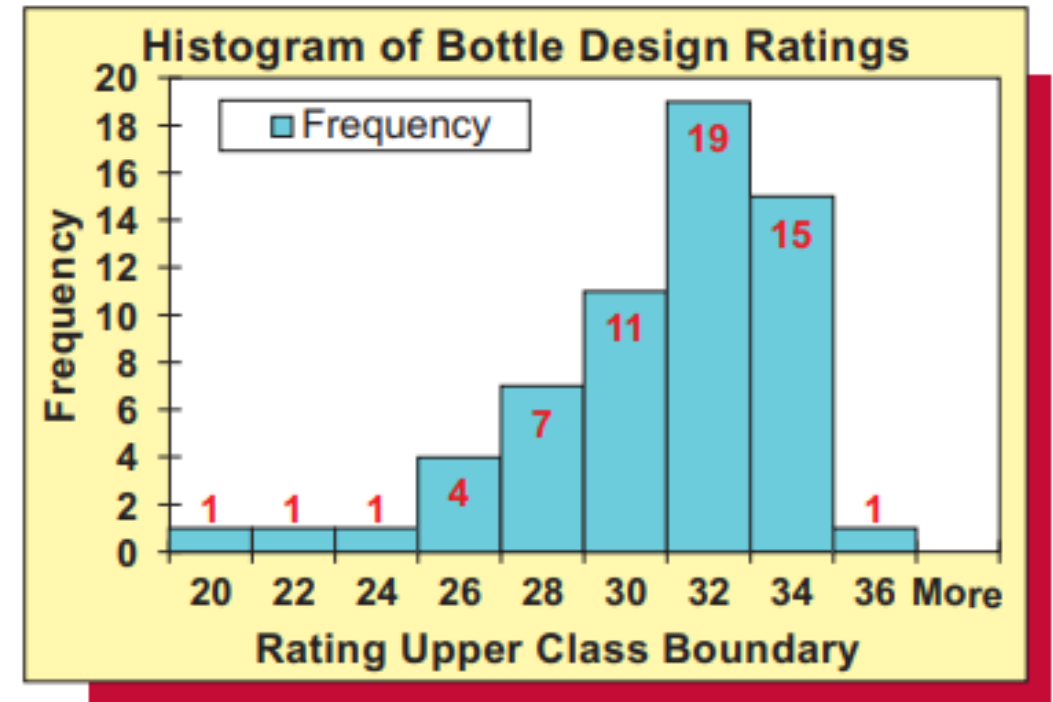


TABLE 1.5 A Sample of Bottle Design Ratings (Composite Scores for a Sample of 60 Shoppers)

34	33	33	29	26	33	28	25	32	33
32	25	27	33	22	27	32	33	32	29
24	30	20	34	31	32	30	35	33	31
32	28	30	31	31	33	29	27	34	31
31	28	33	31	32	28	26	29	32	34
32	30	34	32	30	30	32	31	29	33

Example 3.4: The e-billing Case: Reducing Bill Payment Times

- Figure 3.6 gives a histogram of the 65 payment times of table 2.4
- Mean = 18.108 days
- Median = 17.000 days
- Mode = 16.000 days
- So:
 - . Expect the mean payment time to be 18.108 days
 - . A long payment time would be > 17 days and a short payment time would be < 17 days
 - . The typical payment time is 16 days

FIGURE 3.6 MINITAB Frequency Histogram of the 65 Payment Times

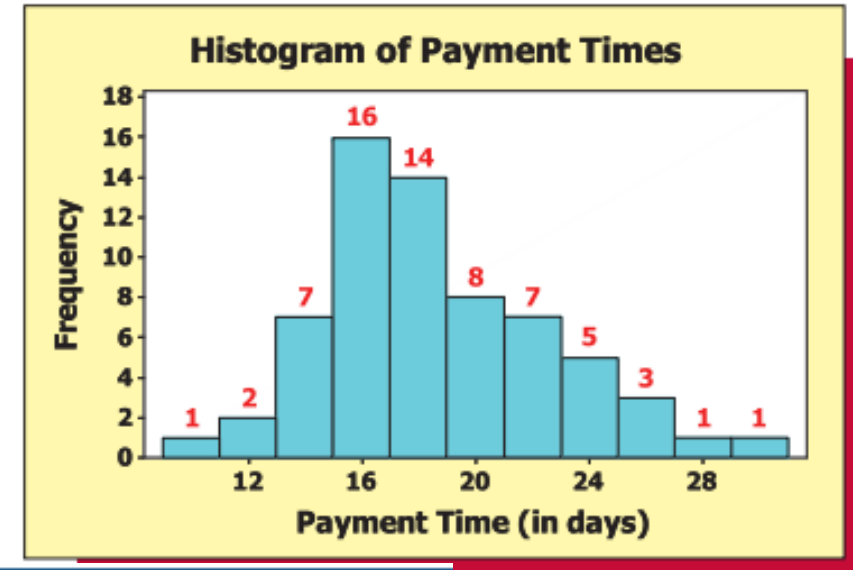


TABLE 2.4 A Sample of Payment Times (in Days) for 65 Randomly Selected Invoices 

22	29	16	15	18	17	12	13	17	16	15
19	17	10	21	15	14	17	18	12	20	14
16	15	16	20	22	14	25	19	23	15	19
18	23	22	16	16	19	13	18	24	24	26
13	18	17	15	24	15	17	14	18	17	21
16	21	25	19	20	27	16	17	16	21	

Example 3.5: The Cell Phone Case: Reducing Cellular Phone Costs

Last month's cellular usages for the 100 randomly selected staff are given in Table 1.4. If we add together the usages, we find that the 100 staff used a total of 46,625 minutes. Furthermore, the total cellular cost incurred by the 100 staff is found to be \$9,317 (this total includes base costs, overage costs, long distance, and roaming). This works out to an average of $\$9,317 / 46,625 = \0.1998 , or 19.98 cents per minute.

TABLE 1.4 A Sample of Cellular Usages (in minutes) for 100 Randomly Selected Employees

 CellUse

75	485	37	547	753	93	897	694	797	477
654	578	504	670	490	225	509	247	597	173
496	553	0	198	507	157	672	296	774	479
0	822	705	814	20	513	546	801	721	273
879	433	420	521	648	41	528	359	367	948
511	704	535	585	341	530	216	512	491	0
542	562	49	505	461	496	241	624	885	259
571	338	503	529	737	444	372	555	290	830
719	120	468	730	853	18	479	144	24	513
482	683	212	418	399	376	323	173	669	611

Questions (at the end of the first session and if time permits)

3.3 Calculate the mean, median, and mode of each of the following populations of numbers:

a 9, 8, 10, 10, 12, 6, 11, 10, 12, 8

b 110, 120, 70, 90, 90, 100, 80, 130, 140

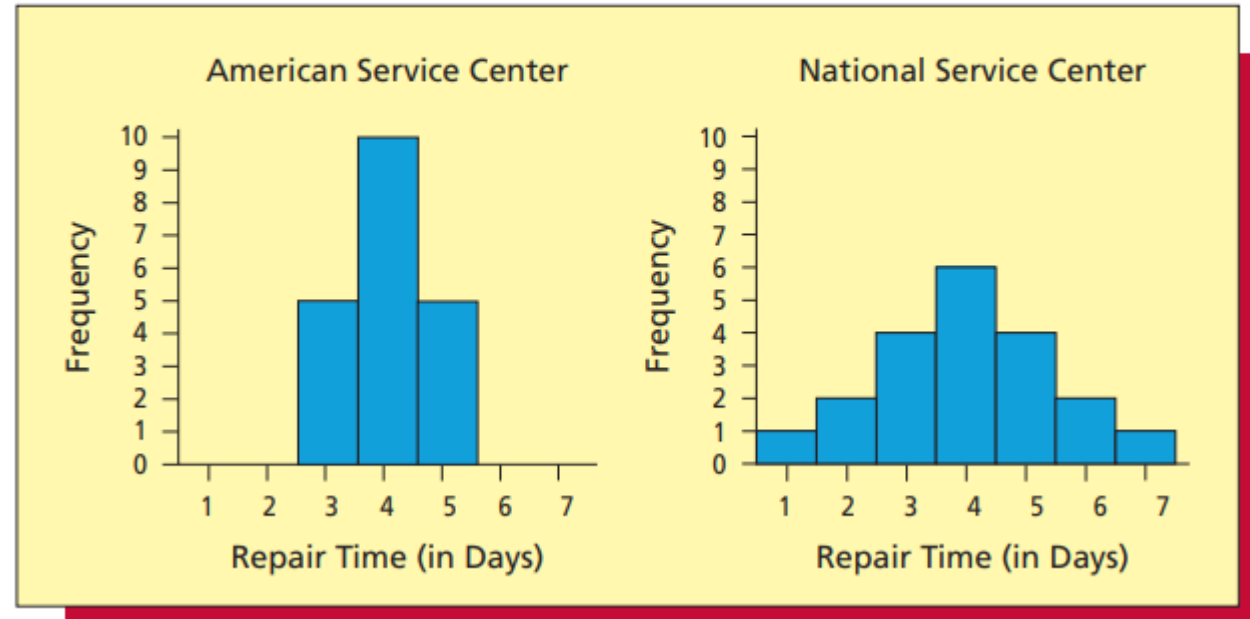
Five minutes break

Please sign your attendance

3.2 Measures of Variation

- In addition to estimating a population's central tendency, it is important to estimate the variation of the population's individual values. For example, Figure 3.13 shows two histograms.
- Because the mean (and median and mode) of each distribution equals 4 days, the measures of central tendency do not indicate any difference between the two centres.
- However, the repair times for the American Service Center are clustered quite closely together, whereas the repair times for the National Service Center are spread farther apart (the repair time might be as little as 1 day, but could also be as long as 7 days). Therefore, we need measures of variation to express how the two distributions differ

FIGURE 3.13 Repair Times for Personal Computers at Two Service Centers



Range: Largest minus the smallest measurement.

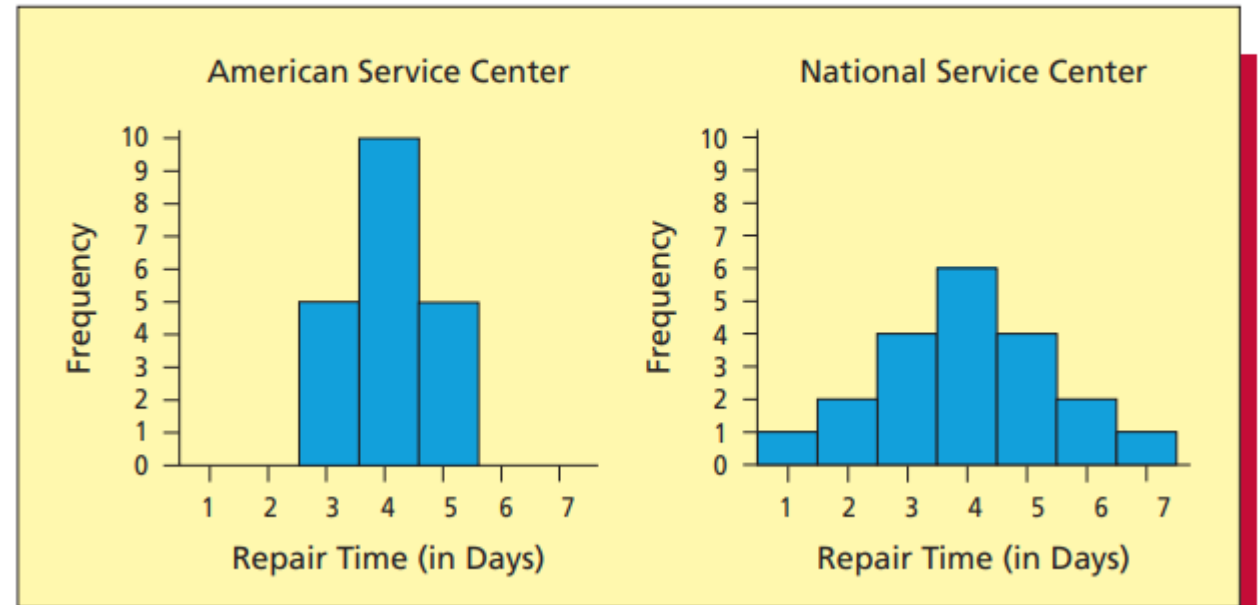
Variance (σ^2): The average of the squared deviations of all the population measurements from the population mean.

Standard Deviation (σ): The square root of the variance.

3.2.1 The Range

- Largest minus smallest
- Measures the interval spanned by all the data
- For American Service Center, largest is 5 and smallest is 3
 - Range is $5 - 3 = 2$ days
- For National Service Center, range is 6

FIGURE 3.13 Repair Times for Personal Computers at Two Service Centers



3.2.2 Variance

- Population of size N

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N} = \frac{(x_1 - \mu)^2 + (x_2 - \mu)^2 + \cdots + (x_N - \mu)^2}{N}$$

- Sample of size n

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1}$$

$$s^2 = \frac{1}{n - 1} \left[\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i \right)^2}{n} \right]$$

We divide by $n - 1$ as this gives an unbiased estimator of the population variance (sometimes this is called Bessel's correction). We will (maybe) learn what an unbiased estimator later in the course.

3.2.3 Standard Deviation

- Population Standard Deviation: $\sigma = \sqrt{\sigma^2}$
- Sample Standard Deviation: $s = \sqrt{s^2}$

Example 3.6: The Car Mileage Case

The first five mileages in Table 3.1: $x_1 = 30.8$, $x_2 = 31.7$, $x_3 = 30.1$, $x_4 = 31.6$, and $x_5 = 32.1$.

$$\bar{x} = 31.26$$

$$\begin{aligned}s^2 &= \frac{\sum_{i=1}^5 (x_i - \bar{x})^2}{5-1} \\&= \frac{(30.8-31.26)^2 + (31.7-31.26)^2 + (30.1-31.26)^2 + (31.6-31.26)^2 + (32.1-31.26)^2}{4} \\&= \frac{2.572}{4} = 0.643 \\s &= \sqrt{s^2} = \sqrt{0.643} = 0.8019\end{aligned}$$

Example 3.1: Car Mileage Case

TABLE 3.1 A Sample of 50 Mileages  **GasMiles**

30.8	30.8	32.1	32.3	32.7
31.7	30.4	31.4	32.7	31.4
30.1	32.5	30.8	31.2	31.8
31.6	30.3	32.8	30.7	31.9
32.1	31.3	31.9	31.7	33.0
33.3	32.1	31.4	31.4	31.5
31.3	32.5	32.4	32.2	31.6
31.0	31.8	31.0	31.5	30.6
32.0	30.5	29.8	31.7	32.3
32.4	30.5	31.1	30.7	31.4

Of course, intuitively, we are likely to obtain more accurate point estimates of the population variance and standard deviation by using all the available sample information. Recall that the mean of all 50 mileages is $\bar{x} = 31.56$. Using this sample mean, it can be verified that

$$s^2 = \frac{\sum_{i=1}^{50} (x_i - \bar{x})^2}{50 - 1} = 0.6363$$

$$s = \sqrt{s^2} = \sqrt{0.6363} = 0.7977$$

The Car Mileage Case: The **Data**

TABLE 3.1 A Sample of 50 Mileages  DS GasMiles

30.8	30.8	32.1	32.3	32.7
31.7	30.4	31.4	32.7	31.4
30.1	32.5	30.8	31.2	31.8
31.6	30.3	32.8	30.7	31.9
32.1	31.3	31.9	31.7	33.0
33.3	32.1	31.4	31.4	31.5
31.3	32.5	32.4	32.2	31.6
31.0	31.8	31.0	31.5	30.6
32.0	30.5	29.8	31.7	32.3
32.4	30.5	31.1	30.7	31.4

Example 3.7: The e-billing Case: Reducing Bill Payment Times

- The sample variance
- The sample standard deviation

$$\sum_{i=1}^{65} x_i = x_1 + x_2 + \cdots + x_{65} = 22 + 19 + \cdots + 26 + 21 = 1,177$$

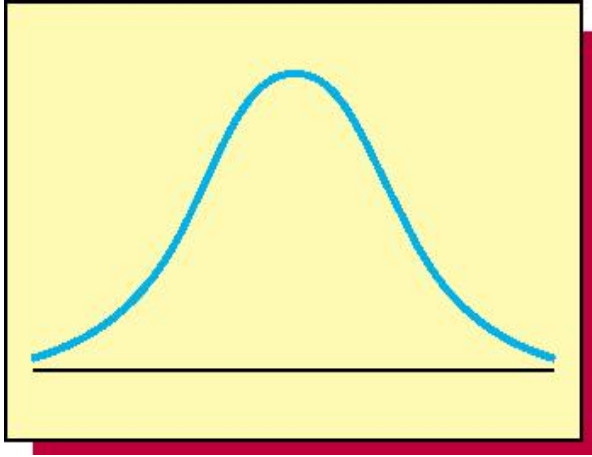
$$\sum_{i=1}^{65} x_i^2 = x_1^2 + x_2^2 + \cdots + x_{65}^2 = (22)^2 + (19)^2 + \cdots + (26)^2 + (21)^2 = 22,317$$

$$\text{Therefore, } s^2 = \frac{1}{(65-1)} \left[22,317 - \frac{(1,177)^2}{65} \right] = 15.69135, \quad s = \sqrt{s^2} = 3.9612 \text{ days}$$

TABLE 2.4 A Sample of Payment Times (in Days) for 65 Randomly Selected Invoices 

22	29	16	15	18	17	12	13	17	16	15
19	17	10	21	15	14	17	18	12	20	14
16	15	16	20	22	14	25	19	23	15	19
18	23	22	16	16	19	13	18	24	24	26
13	18	17	15	24	15	17	14	18	17	21
16	21	25	19	20	27	16	17	16	21	

The Normal Curve



- Symmetrical and bell-shaped curve for a normally distributed population
- If a population is described by a normal curve, we say that the population is **normally distributed**.

The Empirical Rule for Normal Populations

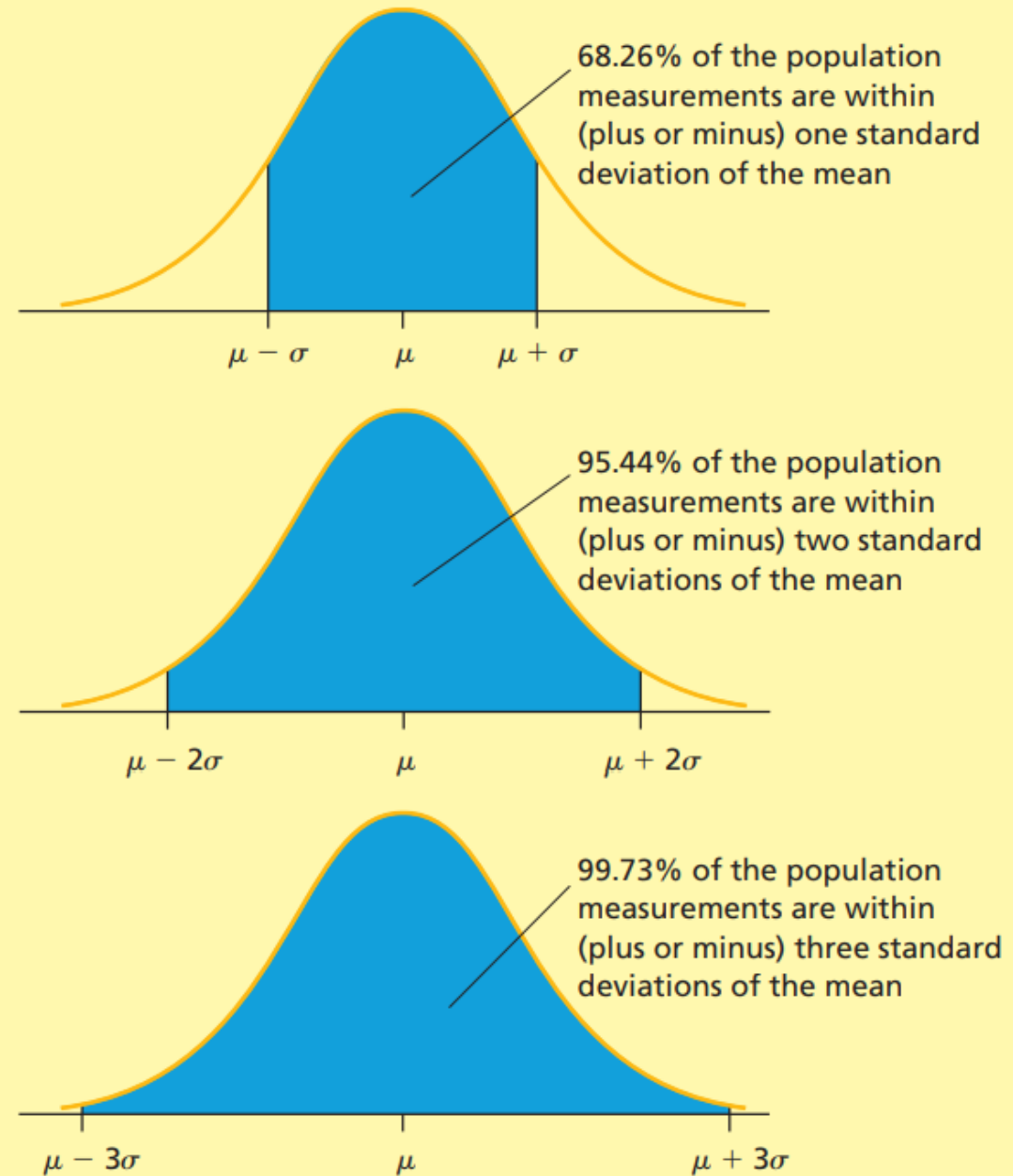
If a population has mean μ and standard deviation σ and is described by a normal curve, then

(1) 68.26% of the population measurements lie within one standard deviation of the mean: $[\mu - \sigma, \mu + \sigma]$

(2) 95.44% of the population measurements lie within two standard deviations of the mean: $[\mu - 2\sigma, \mu + 2\sigma]$

(3) 99.73% of the population measurements lie within three standard deviations of the mean: $[\mu - 3\sigma, \mu + 3\sigma]$

(a) The Empirical Rule



Tolerance Intervals

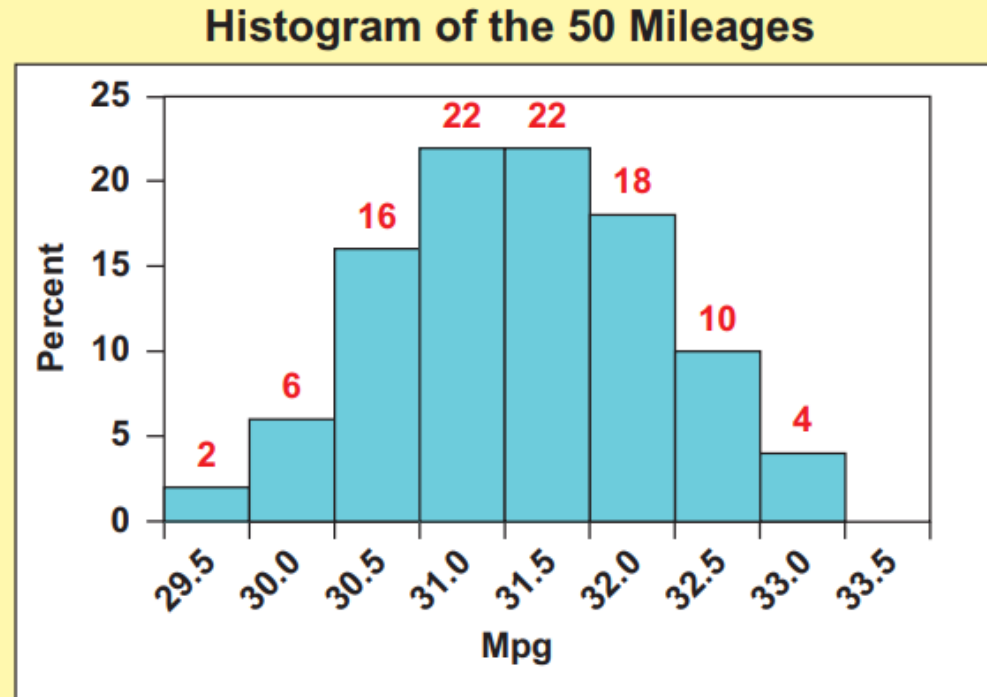
An Interval that contains a specified percentage of the individual measurements in a population is called a **tolerance interval**.

- The one, two, and three standard deviation intervals around μ given in (1), (2) and (3) are tolerance intervals containing, respectively, 68.26 percent, 95.44 percent and 99.73 percent of the measurements in a normally distributed population.
- The *three-sigma interval* $[\mu \pm 3\sigma]$ to be a tolerance interval that contains *almost all* of the measurements in a normally distributed population.

Example 3.8: Estimated Tolerance Intervals in the Car Mileage Case

$$\bar{x} = 31.56$$

$$s = 0.7977$$



[30.8 32.4]

→ Estimated **tolerance interval** for the mileages of 68.26 percent of all individual cars

[30.0 33.2]

→ Estimated **tolerance interval** for the mileages of 95.44 percent of all individual cars

[29.2 34.0]

→ Estimated **tolerance interval** for the mileages of 99.73 percent of all individual cars

The Car Mileage Case

- 68.26% of all individual cars will have mileages in the range

$$[\bar{x} \pm s] = [31.6 \pm 0.8] = [30.8, 32.4] \text{ mpg}$$

- 95.44% of all individual cars will have mileages in the range

$$[\bar{x} \pm 2s] = [31.6 \pm 1.6] = [30.0, 33.2] \text{ mpg}$$

- 99.73% of all individual cars will have mileages in the range

$$[\bar{x} \pm 3s] = [31.6 \pm 2.4] = [29.2, 34.0] \text{ mpg}$$

Because the difference between the upper and lower limits of each estimated tolerance interval is fairly small, we might conclude that the variability of the individual car mileages around the estimated mean mileage of 31.6 mpg is fairly small. Furthermore, the interval $[\bar{x} - 3s, \bar{x} + 3s] = [29.2, 34.0]$ implies that almost any individual car that a customer might purchase will obtain a mileage between 29.2 mpg and 34.0 mpg.

The Car Mileage Case

- 68.26% of all individual cars will have mileages in the range $[\bar{x} \pm s]$

In other words:

- The **probability** of cars having mileages in the range $[\bar{x} \pm s]$ is 0.6826

Next week, we will begin our journey to **Probability**.

Questions (at the end of the session and if time permits)

- 3.19** Consider the following population of five numbers: 5, 8, 10, 12, 15. Calculate the range, variance, and standard deviation of this population.
-

Questions (at the end of the session and if time permits)

- 3.22** In order to control costs, a company wishes to study the amount of money its sales force spends entertaining clients. The following is a random sample of six entertainment expenses (dinner costs for four people) from expense reports submitted by members of the sales force.  DinnerCost

\$157 \$132 \$109 \$145 \$125 \$139

- a** Calculate $\bar{x}$, s^2 , and s for the expense data. In addition, show that the two different formulas for calculating s^2 give the same result.
- b** Assuming that the distribution of entertainment expenses is approximately normally distributed, calculate estimates of tolerance intervals containing 68.26 percent, 95.44 percent, and 99.73 percent of all entertainment expenses by the sales force.
- c** If a member of the sales force submits an entertainment expense (dinner cost for four) of \$190, should this expense be considered unusually high (and possibly worthy of investigation by the company)? Explain your answer.
- d** Compute and interpret the z -score for each of the six entertainment expenses.

Thank you!